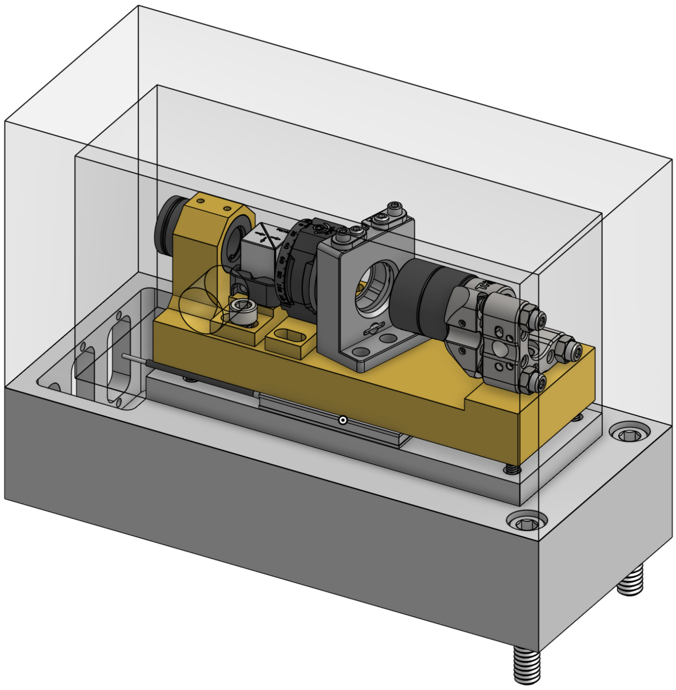
Cateye IF ECDL Setup



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1 Introduction

This handbook provides a guide to the design, alignment, and operation of a cateye interference filter external cavity diode laser (ECDL). The setup combines an ultra-narrow bandpass interference filter and a cateye retro-reflector to achieve robust frequency selectivity and high mechanical stability. Key components such as the feed-forward tuning scheme, optical feedback optimization, and protection circuits are detailed. This document is intended as a practical reference for constructing and maintaining a high-performance ECDL.

1.1 Theoretical Basis for ECDL

The output frequency ν of an ECDL is determined by the combined effects of frequency-dependent gain and loss mechanisms. The laser oscillates at the frequency where the total transmission gain is maximized[2]:

$$T_{\text{total}} = G_D T_D T_{\text{cavity}} T_{\text{filter}} \tag{1}$$

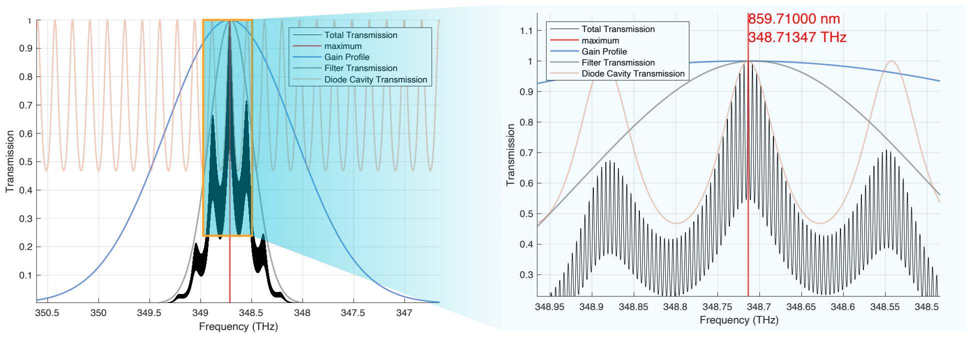


Figure 1: Example simulation result of the total transmission profile of a typical cateye IF ECDL as a function of frequency. The ECDL will lase at the frequency with maximum transmission

Here G_D is the diode's gain profile of the laser diode, T_D is the transmission function of the internal diode cavity. T_{cavity} is the transmission function of the external cavity (e.g., cateye reflector), and T_{filter} is the transmission profile of the interference filter. The diode will lase at the frequency where this product is maximized. Figure 1 shows an example of simulated total transmission profile of a typical ECDL. The transmission function of a Fabry–Perot cavity is:

$$T(\nu) = \frac{1}{1 + F \sin^2\left(\delta(\nu)\right)} \tag{2}$$

where $F = \frac{4r_1r_2}{(1-r_1r_2)^2}$ is the finesse coefficient and $\delta(\nu) = \frac{2\pi n L \nu}{c}$, with r_1, r_2 the reflectivity of the cavity mirrors, L as the cavity length, and n the refractive index.

Tuning the frequency of the ECDL output can be think as shifting the total transmission gain peak, which can be achieved by: rotating the filter angle, changing the temperature, current of the diode, and voltage of the piezo. The wavelength of the transmission peak shifts with filter angle θ given by:

$$\lambda(\theta) = \lambda_0 \sqrt{1 - \left(\frac{\sin(\theta)}{n_{\text{eff}}}\right)^2} \tag{3}$$

where n_{eff} is the index of reflection of the filter's material. Changing the temperature and current of the diode changes G_D and T_D. Increasing current red-shifting the gain peak, which can be thought as increases the diode cavity length via thermal expansion. Increasing the piezo voltage reduce the external cavity length, thus increase the frequency.

1.1.1 Feed-Forward Scheme

Feed-forward in an ECDL is a method to keep the laser frequency stable while tuning. As you adjust the external cavity length via a piezo, the internal diode cavity also needs to shift to stay in resonance and avoid mode hops. Feed-forward works by simultaneously changing the diode current along with the piezo voltage, using a pre-calibrated ratio. This ensures both cavities' transmission peaks remain aligned, allowing smooth, continuous frequency tuning without jumps. This feed-forward scheme can be implemented using either digital or analog methods, as detailed in Sections 2.4 and 2.3.

1.1.2 Why Cateye Configuration?

In a cateye configuration, a lens-mirror combination (Cateye) and a lens-source aligned facing each other, with the mirror/light source at the focal plane/point of the lens. The cateye configuration is chosen over a simple two parallel flat mirror design due to its angular insensitivity, as shown in Figure 2. It reflects any incoming collimated beam back to the source, regardless of their relative angle (provided the beam strikes the Cateye). So it's easy to align and mechanically stable.

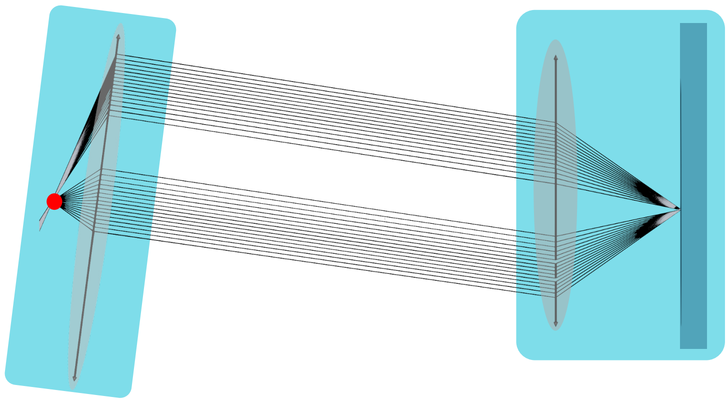


Figure 2: Demonstration of cateye retro-reflector configuration. All the rays emerged from the source goes back to the source

1.1.3 Importance of Feedback Alignment

ECDL performance is highly sensitive to optical feedback. Misalignment reduces feedback efficiency F, which can make the ECDL unstable or multimode. Feedback efficiency is defined by the overlap of emitted and reflected electric fields:

$$F = \frac{1}{R} \left| \iint E_{\mathrm{i}}^{*} E_{\mathrm{r}} dx dy \right|^{2} \tag{4}$$

where R is the coupler reflectivity, which normalizes F so that perfect reflection gives $F = 1$. Small misalignment, either angular (α) or axial (δ), reduce F according to[1]

$$F = \exp\left(- \left(\frac{\alpha \pi \omega_0}{\lambda} \right)^2 \right) \quad \text{and} \quad F = \left(1 + \frac{\delta^2 \lambda^2}{\pi^2 \omega_0^4} \right)^{-1} \tag{5}$$

For a beam waist $\omega_0 \approx 10\ \mu$m, a misalignment of $\alpha = 0.5^\circ$ or $\delta = 0.1$ mm can reduce feedback by 10%. Hence, precise positioning of the cateye reflector is critical. A well-designed ECDL should enable easy adjustment of both axial position and tilt to optimize feedback.

1.2 Design Overview

The cateye IF ECDL comprises two collimating lenses—one to collimate the diode output and the other to focus the beam onto the mirror—a $\lambda/2$ plate and a PBS for precise control of the feedback intensity, an interference filter on a rotational mount for coarse wavelength tuning (see Eq. 3), and a piezo-mounted mirror at the second lens's focal plane that provides fine frequency tuning via the piezo voltage. The self-aligning cateye retro-reflector minimizes sensitivity to mechanical noise and misalignment.

All optical elements are mounted on a 360 Brass block, which in turn rests on an aluminum base with a TEC sandwiched between them. A 1-inch thick acrylic lid encloses the assembly to provide thermal and vibrational stability.

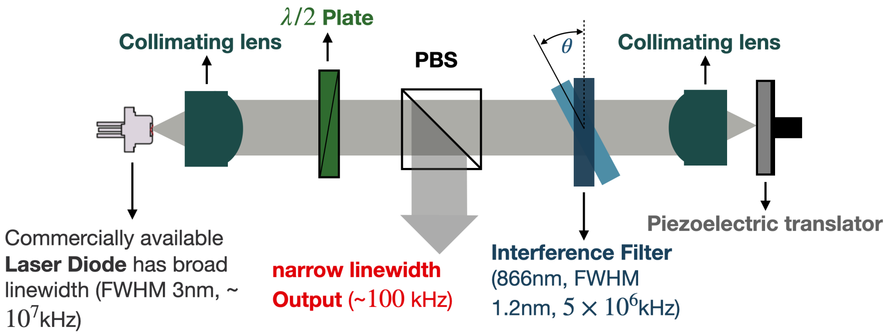


Figure 3: Schematics of Cateye IF ECDL Optics

The total effective feedback in this design can be calculated based on the initial polarization angle of the diode output (measured relative to the vertical axis) denoted by θ, and the orientation of the half-wave plate (HWP), whose fast axis makes an angle ϕ wrt. the vertical axis. Given that a HWP rotates linear polarization by $2(\theta - \phi)$ across the fast axis, and by assuming that the diode keeps lase along its initial polarization direction regardless of the polarization of the feedback (which may not always hold true in practice, as indicated by experimental observations), the effective feedback is then given by the projection of the feedback polarization onto the diode's original polarization direction, i.e., the component of the returning light aligned with the original polarization axis of the diode, one can calculate the effective feedback as

$$I_{\text{eff}} = I_0 \frac{1}{4}(\cos(2\theta) - \cos(4\phi))^2 \tag{6}$$

This result indicates that the effective feedback depends sensitively on the polarization of the input laser diode. The analysis assumes the diode's output polarization remains fixed; however, in reality, the situation can be more complex. If the laser diode's gain is not sharply peaked (i.e., not a delta-function) with respect to polarization, the output polarization may shift in response to strong injected feedback with a different polarization. In such cases, the diode may favor lasing along a new polarization axis that maximizes the net gain, including contributions from both internal and external feedback. This feedback-induced polarization instability introduces nonlinear behavior and complicates the analysis significantly, potentially requiring a full model of the polarization-dependent gain dynamics. Fortunately, experimental experience shows that a stable ECDL can still be achieved even when polarization-dependent gain effects are present.

2 Circuits

2.1 Laser Diode Protection Circuit

To ensure the safe and stable operation of the laser diode (LD), a protection circuit is implemented, as illustrated in Figure 4. This circuit suppresses voltage transients, mitigates electromagnetic interference (EMI), and limits current surges. The circuit can be divided into three functional blocks: **protection diodes, low-pass filter, and current-limiting resistor**.

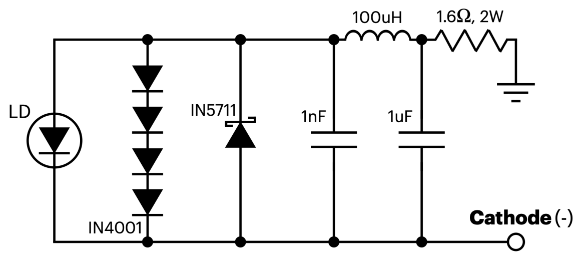


Figure 4: Laser diode protection circuit

Transient and Reverse Polarity Protection

- **1N4001 Diodes (×4)**: These general-purpose silicon diodes are connected in series and placed in parallel with the laser diode to provide clamping protection against voltage transients, particularly electrostatic discharge (ESD) or switching spikes. Each diode has a forward voltage drop of approximately 0.7 V. When forward biased, the total voltage drop across the four diodes is approximately 2.8 V, which effectively clamps the voltage across the laser diode and prevents it from exceeding a safe threshold. This configuration ensures that any transient voltage above this limit is diverted through the diodes rather than damaging the LD.

- **1N5711 Schottky Diode**: Placed in parallel with the LD in reverse orientation, this diode protects against reverse voltage. Since laser diodes are extremely sensitive to reverse bias, even small reverse voltages can cause permanent damage. The 1N5711 has a low forward voltage (∼0.4 V), allowing it to conduct and clamp quickly if a reverse voltage appears.

Low-Pass Filtering Network

To suppress noise from the power supply and reduce high-frequency EMI coupling into the LD, a low-pass filter is implemented using a combination of inductors and capacitors:

- $100\mu H$ **Inductor**: This component provides high impedance to high-frequency signals, effectively attenuating them. It serves as a choke to block rapid voltage transients and forms the central component of a pi-filter.

- $1nF$ **and** $1\mu F$ **Capacitors**: These capacitors are placed to ground before and after the inductor. They provide a low-impedance path for high-frequency noise, shunting it away from the LD.

  The combined structure (C–L–C) forms a second-order passive low-pass filter. The cutoff frequency f_c for such a filter is given by:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

where C is the equivalent capacitance of the two capacitors. Because the capacitors are separated by the inductor, the dominant attenuation is determined by the smaller capacitor (1 nF) at high frequencies. Using $L = 100$ μH and $C = 1$ nF:

$$f_c \approx \frac{1}{2\pi\sqrt{(100 \times 10^{-6})(1 \times 10^{-9})}} \approx 500 \text{ kHz}$$

This ensures effective attenuation of switching noise and RF interference above the kHz range.

Current-Limiting Resistor

- **1.6Ω, $2W$ Resistor**: This resistor limits inrush current and provides protection against short-duration voltage spikes that might bypass other protection mechanisms. The power rating of $2W$ allows it to handle transient pulses without damage.

  The voltage drop across the resistor during normal operation can be calculated using Ohm's Law:

  $$V = IR$$

  For example, if the LD operates at 100 mA:

  $$V = 0.1 \text{ A} \times 1.6 \ \Omega = 0.16 \text{ V}$$

  The resistor thus causes a small voltage drop while offering valuable current limiting protection.

2.2 Piezo Protection Circuit

To protect the piezoelectric actuator from high-frequency noise and voltage spikes, a simple RC low-pass filter is employed at its input, shown in Figure 5. This configuration helps prevent unwanted oscillations or electrical interference from affecting the piezo response.

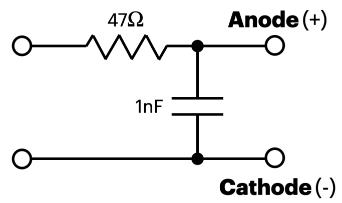


Figure 5: Piezo protection circuit

The capacitor and resistor forms a RC low-pass filter. The cutoff frequency f_c of this RC low-pass filter is calculated as:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 47\Omega \times 1\text{nF}} \approx 3.4\text{MHz}$$

This means signals above $\sim$3.4 MHz are significantly attenuated.

2.3 Digital Feedforward Circuit

Adjusting either the diode current or the piezo voltage independently yields a relatively limited mode-hop-free tuning range (MHFTR), typically around 1.5 GHz. A straightforward method to extend the MHFTR is to implement feedforward control, that is, to adjust the current and piezo voltage simultaneously with a certain ratio. By compensating the frequency shift induced by one tuning mechanism with the other, the laser can be kept on the same longitudinal mode over a broader frequency range. This technique effectively increases the usable, continuous tuning range of the ECDL without mode hops. This technique can be implemented through either digital or analog means, though digital implementations may introduce additional noise due to DAC resolution and sampling artifacts. In this section, we first describe the digital feedforward scheme, where both the diode current and piezo voltage are controlled via a computer through the DAQ.

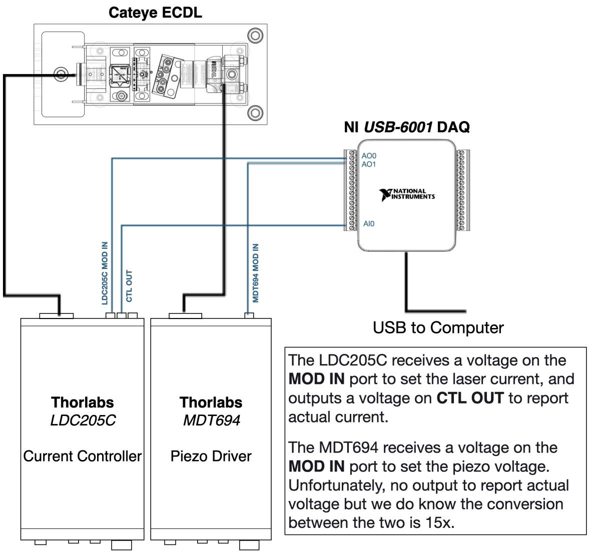


Figure 6: Schematics for the digital feedforward circuit

Figure 6 shows the schematics of a digital feedforward circuit. The setup uses a National Instruments USB-6001 DAQ to interface with both the Thorlabs LDC205C current controller and the MDT694 piezo driver, enabling complete control of the laser diode current and piezo voltage via a computer. The LDC205C receives a control voltage from DAQ output AO0 at its MOD IN port, which sets the laser diode current. The actual output current is reported via the CTL OUT port of the LDC205C, which is connected to DAQ input AI0 for real-time monitoring. The MDT694 receives a control voltage from DAQ output AO1 at its MOD IN port to set the voltage applied to the piezo. The MDT694 does not provide a feedback voltage for monitoring, but the voltage gain is known ($15\times$), so the piezo voltage can be inferred. The DAQ is connected to a Python script via USB, which also communicates with a HighFinesse WS/7-8395 wavemeter to read the laser frequency in real time. This configuration allows for precise mapping of current vs. frequency and voltage vs. frequency, which can be very helpful when finding the optimal feedforward ratio (the ratio between input voltage from piezo driver and the modulation voltage send to the current controller) as this ratio can be easily adjusted with a minimal adjustment to the code.

2.3.1 Results

Laser frequency as a function of laser diode current and piezo voltage is shown in Figure 7. The y-axis in all plots represents the frequency offset Δf from a reference value of 348.697 THz, as measured by the HighFinesse wavemeter.

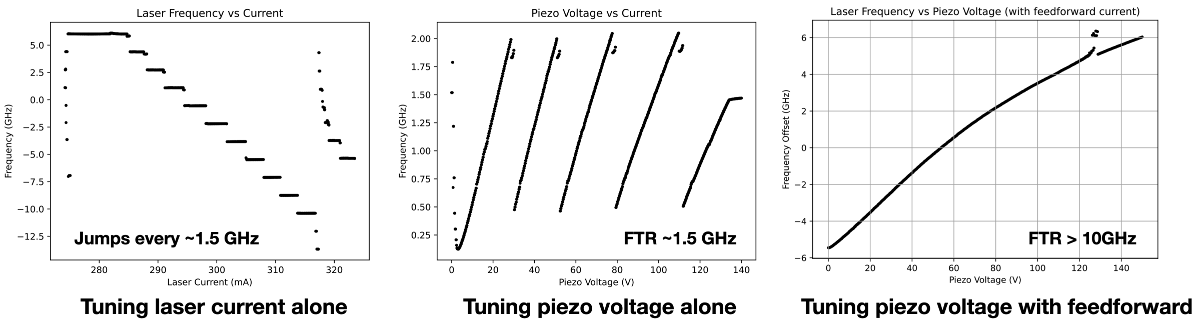


Figure 7: Laser output frequency as scanning laser current alone (left), scanning piezo voltage alone (middle), and scanning piezo voltage with feedforward (right)

Sweeping the laser current alone (left) gives a discrete jumps in frequency with interval aroudn 1.5 GHz, with clear mode hops occurring at regular intervals; scanning the piezo voltage alone (middle) yields a mode-hop-free tuning range (MHFTR) of approximately 1.5 GHz; simultaneous sweeping piezo voltage while adjusting the diode current proportionally (right) brings the MHFTR to above 10GHz. A larger MHFTR could be achieved in principle by using a piezo with longer displacement range, but in my case, replacing to a longer piezo with 3 times displacement range doesn't increase the range considerably (15GHz).

2.3.2 Notes and Tips

1. **Start with Low Current:** To prevent potential damage from voltage spikes at startup, always begin with a low laser diode current (typically 50–100mA). Alternatively, unplug the diode before enabling DAQ control. Ensure that the current limit (`ILIM`) is properly configured on the LDC205C front panel.

2. **Stabilization Delays:** When automating wavelength scans or feedback loops, include small delays (e.g., `time.sleep(0.01)`) between DAQ updates to allow the wavemeter and laser to stabilize.

3. **Confirming DAQ Device Name:** Use the following Python snippet to identify the correct DAQ device name (e.g., `Dev1`), which is required for DAQ channel assignments:

   ```python
   import nidaqmx.system
   print([d.name for d in nidaqmx.system.System.local().devices])
   ```

4. **HighFinesse Wavemeter Access:** When reading the laser frequency using the HighFinesse WS/7 wavemeter in Python, use the function `getFrequencyNum(x, 0)` with channel x = 3 or 4, depending on your wavemeter's configuration.

2.4 Analogue Feedforward Circuit

Unlike the digital feedforward method (implemented using Python and a DAQ), the analog approach introduces no quantization noise, making it inherently quieter and more deterministic. The core functionality of the circuit shown in Figure 8 is to linearly scale an input voltage in the range of 0–150V down to a smaller output range (e.g., 0–3.4V). This is achieved using an inverting amplifier configuration, where the gain is set by a resistor network (typically a potentiometer), allowing fine adjustment of the scaling factor to match the desired current modulation response.

$$V_{\text{out}} = kV_{\text{in}}$$

where k is set by a 500Ω potentiometer (e.g., a Bournes CT-23-6A dial resistor). The laser's piezo control voltage $V_{\text{pzt}} \in [0, 150\,\text{V}]$ (from the Thorlabs MDT693A piezo driver) is used as the input V_{in} to a scaling circuit. The output V_{out} is then fed into the MOD IN R2 pin on the LDC205C current controller as in the digital circuit. This voltage adjusts the laser current with a modulation range of ±0.5A corresponding to ±10V input (i.e., 50mA/V sensitivity).

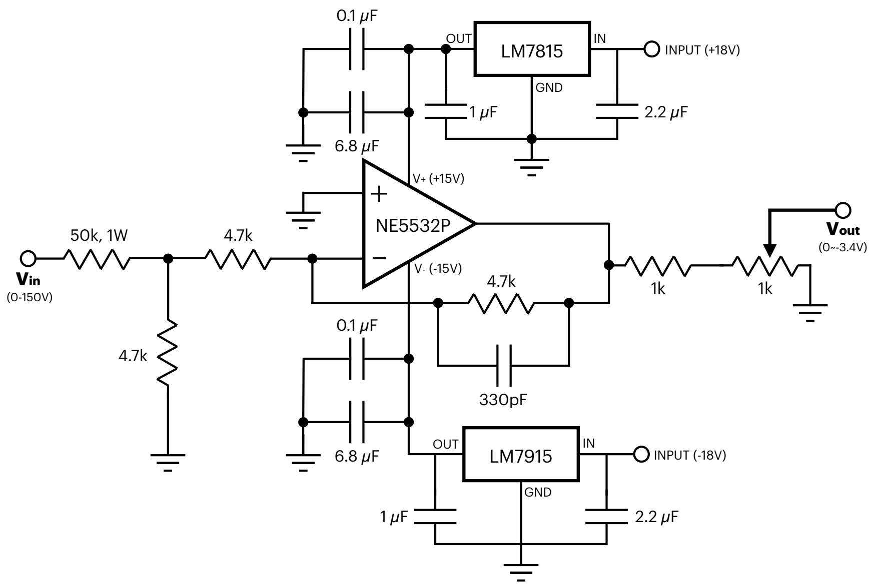


Figure 8: Schematics for the analogue feedforward electronic circuit

2.4.1 Determining the Feedforward Ratio

To find the appropriate scaling factor k, we first characterize the system. Based on the result presented using digital circuit, the piezo frequency sensitivity (without feedforward) is

$$\sigma_V = \frac{\Delta f}{\Delta V_{\text{pzt}}} \approx \frac{1.5\,\text{GHz}}{25\,\text{V}} = 60\,\text{MHz/V} \tag{7}$$

The current sensitivity is measured by identifying the frequency jump between two cavity modes and dividing by the change in diode current needed to cause the jump:

$$\sigma_I = \frac{\Delta f}{\Delta I} \approx -431\,\text{MHz/mA} \tag{8}$$

To compensate for a piezo-induced frequency shift using current tuning, we require:

$$\frac{\Delta I}{\Delta V_{\text{pzt}}} = \frac{\sigma_V}{\sigma_I} \approx \frac{60}{-431} = -0.139\,\text{mA/V} \tag{9}$$

So, for the full 150V piezo sweep:

$$\Delta I = -0.139\,\text{mA/V} \times 150\,\text{V} = -20.9\,\text{mA} \tag{10}$$

This corresponds to a maximum modulation input of:

$$V_{\text{MOD}}^{max} = \frac{-20.9\,\text{mA}}{50\,\text{mA/V}} = -0.418\,\text{V} \tag{11}$$

The potentiometer is then set to $V_{\mathrm{MOD}}^{max}/V_{\mathrm{out}}^{max}$ of its maximum value, which may be accomplished with relatively good precision by using a 10-turn potentiometer and dial scale (Bournes CT-23-6A or similar). V_{out}^{max} is the voltage at cross at the left of two 1k parallel resistor which is calculated to be -6.73V in this case. Therefore,

$$V_{\mathrm{MOD}}^{max}/V_{\mathrm{out}}^{max} = \frac{0.418}{6.73} = 0.062 \tag{12}$$

2.4.2 Result

Below shows the piezo voltage vs. output frequency with analogue feedforward scheme.

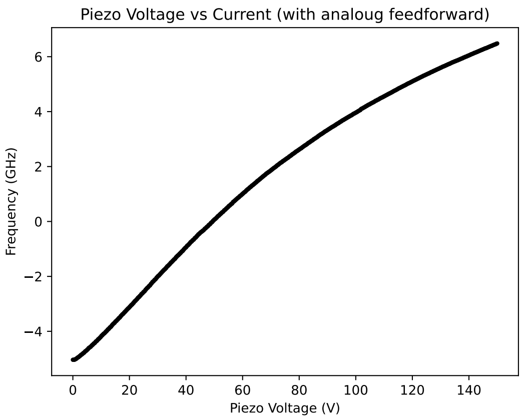


Figure 9: Scanning piezo voltage vs. laser output frequency with analogue feedforward circuit

3 Alignment Procedure

1. **Collimating the Laser Diode:** Insert the laser diode into the collimation tube and mount it onto the brass base. Remove all other components from the base, as depicted in Figure 10. Turn on the diode current (with a protection circuit) until a laser output is visible. Collimate the beam by adjusting the aspherical collimation lens with a spanner wrench to minimize spot size at the greatest possible distance (∼3m would be sufficient). Ensure the lens is securely fixed within the collimation tube when optimal collimation is achieved. If optimal collimation cannot be obtained within the available adjustment range for the lens, consider using a collimating lens with a different focal length.

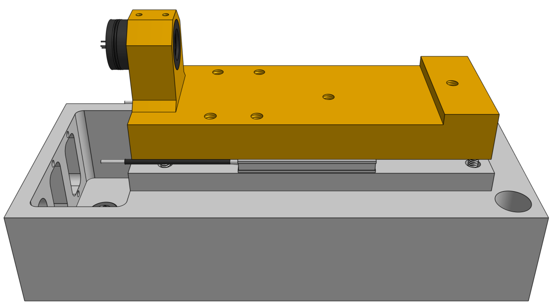


Figure 10: Collimating the laser diode

2. **Collimating the Cateye Reflector:** Reduce the diode current to just above threshold to prevent diode damage from excessive feedback. Assemble the cateye reflector (lens tube + Polaris threaded mirror mount + piezo + collimating lens) and mount it along with the beam splitter (BS) holder onto the brass base, as shown in Figure 11. Position the BS holder as close to the diode collimation tube as possible so the lens can be easily adjustments with a spanner wrench. It is recommended to choose the smallest available focal-length lens for enough space for later placing the filter. Temporarily put a non-polarizing BS cube on to the holder. Roughly align the cateye reflector so the return beam hit onto the BS and partially reflected to the side. Tune the beam collimation by adjusting the reflector's lens position, by looking at the reflected beam.

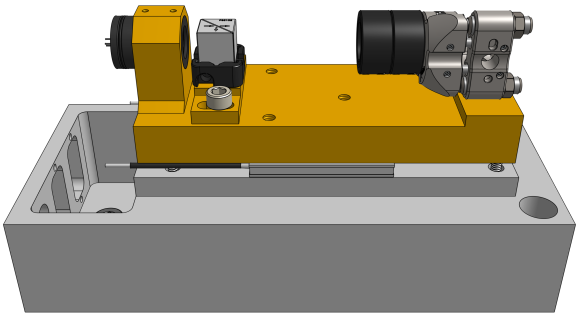


Figure 11: Collimating the cateye reflector

3. **Aligning the Cateye Reflector for Maximum Feedback:** Assemble all components on the brass base as illustrated in Figure 12. Reinstall the PBS, ensuring the filter is angled for maximum transmitted intensity. (Incorporating the filter at this stage accounts for the beam path shift caused by the filter). Set the diode current slightly above threshold current (∼1 mW output power) and place a power meter at the output. Proper feedback alignment is indicated by a significant decrease in output power (or shift in frequency if connected to a wavemeter) upon blocking the feedback beam (cateye reflector). If not observed, refine alignment by adjusting the cateye reflector's pitch and roll or rotate the half-wave plate (HWP). (If necessary, use an card with a small aperture for coarse alignment). Then, rotate the HWP until most of the power goes to the output to prevent strong feedback broke the diode.

   Note: During this process, when turn up the laser diode's current, a counterintuitive increase in output power may be observed when the feedback path is blocked. This effect may arise due to a mismatch in polarization between the incident and feedback beams. Although the laser diode typically exhibits strong polarization-dependent gain, sufficiently strong feedback can alter the polarization state of the emitted beam. This change in polarization can change the power distribution at the output. Thus, this phenomenon is likely diode-dependent and may vary significantly across different diodes.

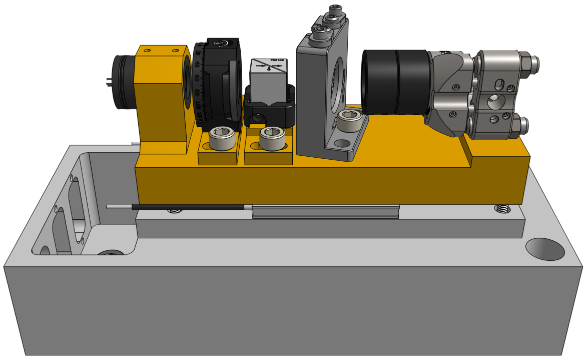


Figure 12: Cateye reflector alignment for maximum feedback.

4. **Adjusting the Half-Wave Plate (HWP) Angle:** Set up downstream optics such as an isolator, coupling mirrors, and fiber-coupling components, etc. Connect the fiber output to a wavemeter and Fabry–Perot cavity. Slowly rotate the HWP angle until a stable single-mode is observed. An indicator of a good optical feedback is a shift in laser frequency when the cateye reflector is blocked, or varying the laser diode current (without feedforward) should result in discrete frequency steps. If neither of these behaviors is observed, repeat the previous alignment procedure.

5. **Tuning Laser Frequency:** Set the diode current to its recommended operating current. (Remember to set the current limit on the current controller to be slightly smaller than the max current rating on the diode's data sheet). Temporarily block the cateye reflector to determine the diode's free-running wavelength. Adjust ECDL temperature until the free-running frequency of the diode is closest to the target frequency. Then, unblock the cateye reflector and tune the filter angle for maximum filter transmission. Adjustments the filter angle to coarse tuning the frequency. One should be able to see the frequency to jump in smallest unit around 3GHz. Fine tune the diode current until the frequency is within 1GHz to the target frequency. Final precise tuning to the desired frequency is achieved through tuning the piezo voltage.

6. **Implementing Feedforward:** Detailed instructions on the analogue feedforward circuit and its implementation are provided in Section 2.4.

3.1 Troubleshooting

1. **Flat Fabry-Perot Cavity Signal:** When coupling the laser into the cavity to find a single-mode, you may occasionally observe a flat, noise-like signal, even though it's clear the laser intensity hitting on the PD is non-zero. This is likely due to excessive optical feedback. Try rotating the laser diode's input polarization/angle, as a simple calculation shows that the input polarization influences the effective feedback. This can be easily seen as the feedback from the cateye is always vertically polarized. If the problem persists, introduce a slight intentional misalignment of the cateye to further lower the feedback intensity.

2. **Temperature Fluctuation:** Achieving stable PID temperature control has been challenging, probably because the thermistor was originally 0.6 in away from the TEC which makes the response time large. We used Brass 360 in the original design, whose thermal conductivity is moderate; an aluminum base should perform better. In the revised design the thermistor hole is closer to the TEC, though this version has not yet been tested. Qian tried with aluminum and observed that when placing the thermistor near the diode

(farther from the TEC) leads to temperature oscillations, whereas placing it near the TEC enables stable control after tuning.

A Part List

The following list includes only the components required to construct the ECDL itself. It does not cover downstream optics such as optical isolators, fiber coupling assemblies, or diagnostics, as well as the circuits.

A.1 Hardware (Wavelength Independent)

Item	Part Number	Supplier	Quant.	Unit Cost
Laser Diode Socket	S7060R	Thorlabs	1	\$7.49
Lens Cell Adapter	S05TM09	Thorlabs	1	\$23.68
Lens Tube with External Thread	SM05L05	Thorlabs	1	\$16.32
Filter Mount	SM05L05	Newport	1	\$209.00
Beamsplitter Platform	BSH10	Thorlabs	1	\$49.85
Threaded Mirror Mount	POLARIS-K05T6	Thorlabs	1	\$208.47
HWP Rotation Mount	RSP05	Thorlabs	1	\$89.72
Thermoelectric	CP60440	DigiKey	1	\$24.22
Thermistor	TH10K	Thorlabs	1	\$5.32
Piezo Ring Chip	PA44LEW	Thorlabs	1	\$95.96
Discrete Piezo Ring Stack	PK44LA2P2	Thorlabs	1	\$155.77
Acrylic Sheet	8560K279	McMaster	1	\$21.85
Custom Al and Brass Parts		Xometry	1	∼\$400.00

A.2 Optical Parts (Wavelength Dependent)

Note: The part list provided in this section is wavelength dependent. Please select components appropriate for your target wavelength. The items listed here are intended as an example for a ECDL operating at 860nm. At this wavelength, some parts requires some efforts to get, such as the laser diode and filters.

Item	Part Number	Supplier	Quant.	Unit Cost
Laser Diode	QLD-850-350S	Qphotonics	1	\$240.00
Interference Filter	866-1.2 OD6	Alluxa	1	\$3,325.00
Collimating Lens, f = 3.1 mm	C330TMD-B	Thorlabs	1	\$104.03
Collimating Lens, f = 11.00 mm	A220TM-B	Thorlabs	1	\$105.82
PBS Cube	PBS102	Thorlabs	1	\$237.69
HWP	WPH05ME-850	Thorlabs	1	\$311.54
Ø7.0 mm Thin Mirror	BB03-E03	Thorlabs	1	\$49.85
BS Cube	BS011	Thorlabs	1	\$219.89
Collimation Tube for diode	LT230P-B	Thorlabs	1	\$193.67

A.3 Controllers

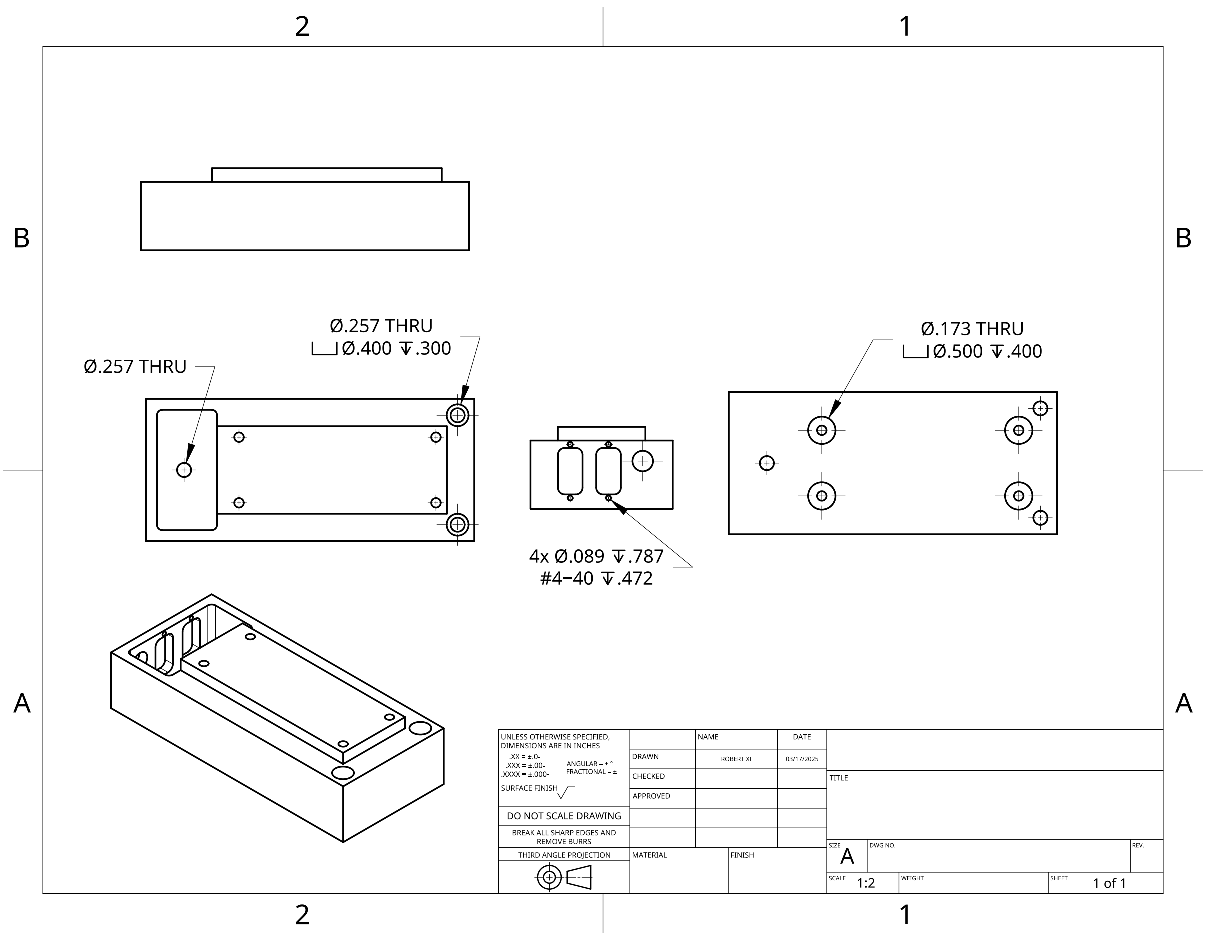
Item	Part Number	Supplier	Quant.	Unit Cost
Temperature Controller	TEC200C	Thorlabs	1	$1,255.82
LD Current Controller	LDC205C	Thorlabs	1	$1,295.03
Piezo Driver	MDT694	Thorlabs	1	
DC Power Supply	E3620A	KeySight	1	$1,301.00
DAQ	USB-6001	National Instruments	1	$270.40

A.4 Other

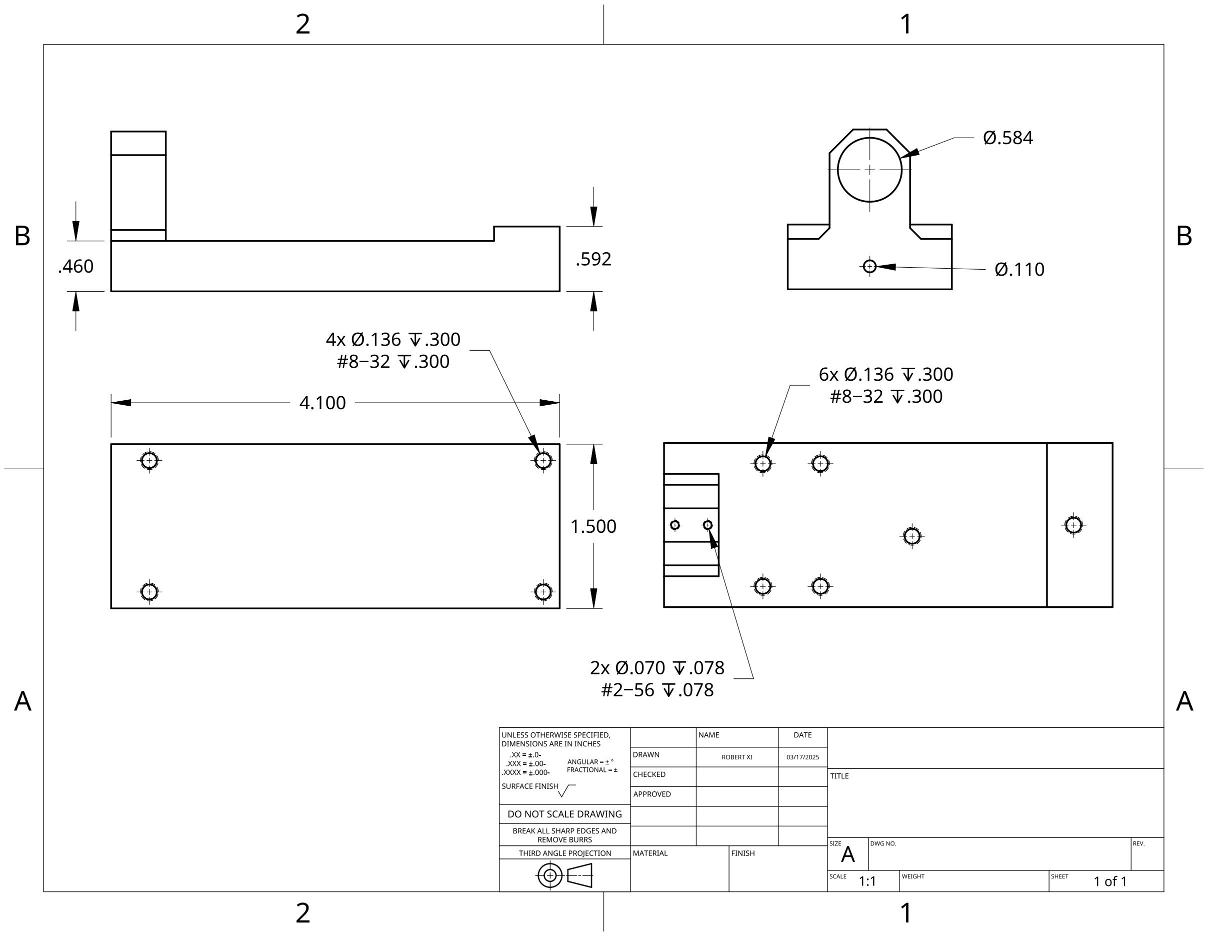
Item	Part Number	Supplier	Quant.	Unit Cost
Cement for Acrylic	7517A1	McMaster	1	$12.30
5 mins Thermal Adhesive	ASTA-7G	Acrtic Silver	1	$20.00

B Mechanical Drawings

Aluminum Base



Brass Base



References

[1] X. Baillard, A. Gauguet, S. Bize, P. Lemonde, Ph. Laurent, A. Clairon, and P. Rosenbusch. Interference-filter-stabilized external-cavity diode lasers. *Optics Communications*, 266(2):609–613, October 2006.

[2] Daniel J. Thompson and Robert E. Scholten. Narrow linewidth tunable external cavity diode laser using wide bandwidth filter. *Review of Scientific Instruments*, 83(2):023107, 02 2012.